MIL-A-20515
4 December 1951

MILITARY SPECIFICATION

ACTIVATOR, M1
LOADING, ASSEMBLING, AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army AXS-1093 (Rev. 1)
7 March 1944

This specification consists of this cover sheet, Specification AXS-1093 (Rev. 1), dated 7 March 1944.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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ACTIVATOR M1
LOADING, ASSEMBLING, AND PACKING

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department Drawing, 73-9-16 and all drawings and specifications listed or referred to thereon, pertaining to loading, assembling and packing, shall form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPE.

B-1. This specification covers one type of activator.

C. MATERIAL AND WORKMANSHIP.

C-1. Materials and parts shall be in accordance with applicable specifications and drawings.

C-2. All parts shall be free from chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. The activator shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. No activator shall fail to detonate with high order when tested with firing device from an accepted lot of a type which is standard for use with this activator (see par. F-6).

E-3. Determination of specific gravity of tetryl pellet. The specific gravity of the tetryl pellet shall be determined by using a method having an error of less than 1 in 100 parts.

F. METHODS OF SAMPLING, INSPECTION AND TESTS.

F-1. Inspection. Inspection shall be as specified in United States Army General Specification for Ammunition Except Small Arms Ammunition, No. 50-0-1.

F-2. Size of lots. Twenty thousand maximum. A smaller quantity may constitute a lot at the discretion of the inspector. Two thousand plus or minus 500 shall constitute the first lot manufactured on any contract, except when a new contract immediately follows the manufacture of satisfactory material of the same character.

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F-3. Sampling. All samples shall be selected by the inspector. Twenty activators shall be selected from each lot for test as specified in paragraph F-6. Sampling of component materials is prescribed in paragraphs covering tests of the materials.

F-4. Determination of specific gravity of tetryl pellet. Ten sample tetryl pellets selected by the inspector from each lot before assembling the activators may be subjected to the following test:

F-4a. Select pellets which are as free of surface cavitation and broken edge as possible. Determine the height and diameter of each pellet by means of a micrometer to within $\pm$ .001 inch. From these measurements determine the volume of each pellet in cubic inches and convert this value to volume in cubic centimeters. This may be accomplished as follows:

Volume (cu. inches) x 16.39 = Volume (c.c.)

Weigh each pellet in grams to $\pm$ .01 gram and record the weight of pellets against the respective volumes. The specific gravity of the pellet is calculated as follows:

$$\text{Specific Gravity} = \frac{W}{V \times A}$$

where

W = Weight of pellet in grams

V = Volume of pellet in cu. centimeters

A = Density of water at 4°C. = 1 gram/cc.

F-4b. Other tests may be used to determine specific gravity of tetryl pellet provided the tests meet the requirement specified in paragraph E-3.

F-5. Determination of graphite content of tetryl pellets. The test shall be conducted as follows:

F-5a. Remove the outer surface from the tetryl pellets that have been used in the specific gravity determination to obtain a composite sample of about 5 grams from the interior of the tetryl pellets. Weigh the sample accurately and transfer to a 250 cc. beaker. Add 100 cc. of acetone and heat to dissolve the tetryl. Filter the solution through a tared filtering crucible and wash the residue in the crucible with two 25 cc. portions of acetone. Dry the crucible at 100°C. for 1 hour; cool in a desiccator, and weigh. Calculate the increase in weight as percentage of graphite.

F-5b. Retest. A retest, using double the number of samples used in the original test, may be made, if requested by the contractor, provided the analysis indicates that the graphite content does not exceed the amount prescribed on the drawing by more than 0.25 percent.

F-6. Test of assembled activators. The test shall be conducted as follows: Assemble twenty of the sample activators with firing devices from an accepted lot of a type which is standard for use with this activator, and test by functioning the firing device in the usual manner. All sample activators shall function with high-order detonation. Consider high-order detonation to be evidenced by the absence of undetonated explosive and by characteristic loud report and absence of white smoke.

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F-5a. Retest. A retest, using double the number of samples may be made if requested by the contractor, provided that not more than one activator does not function with a high-order detonation when tested with a firing device from an accepted lot. No failure shall occur in the retest.

G. PACKAGING, PACKING, AND MARKING.

G-1. Packing.

G-1a. As specified on the drawing.

G-1b. Data Card. Data Cards, 5 x 8 inches, made of commercial manila tag board approximately 200 pounds ream weight, shall be prepared for each lot of accepted activators in accordance with the form specified below. Two of these cards completely filled out for each lot shall accompany each shipping ticket and six cards completely filled out shall be furnished the inspector.

AMMUNITION DATA CARD

Kind: Activator, M1 Ammunition Lot No._____

Quantity in Lot_____

Drawing No. _____Rev. _____ Spec. No._____Rev._____

Parts Manufactured by_____ Lot No. _____

Loaded and Assembled by_____ Date_____

Contract No._____ Order No. _____

Packed _____ in a box.

Remarks:_____

COMPONENTS

Parts	Lot No.	Manufacturer	Date of Manufacture	Loaded	Date of Loading

Certified to by_____, Inspector,
(Print here name of inspector, district, arsenal, or loading plant, and sub-district where applicable.)

G-2. Marking. The marking for shipment shall be as specified on the drawing.

H. NOTES.

H-1. The contract should list the materials to be furnished by the loading and assembling contractor.

H-2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

Note: Copies of this specification may be obtained from the Office, Chief of Ordnance, Washington, D. C.

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